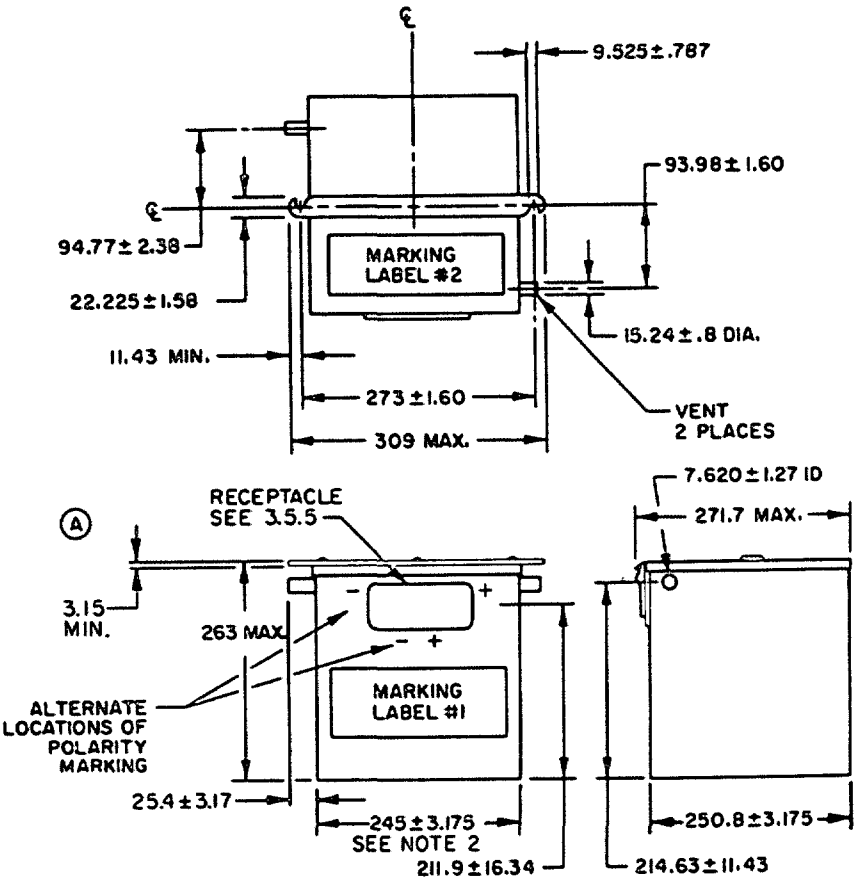
MILITARY SPECIFICATION SHEET

Ⓐ BATTERY, STORAGE, AIRCRAFT, MEDIUM-RATE, TYPE 1, MAINTENANCE FREE, 24-VOLT, 30 AMPERE-HOUR

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the storage battery described herein shall consist of this document and the latest issue of Specification MIL-B-8565.




NOTES: 1. All dimensions are in millimeters.
2. Allow additional 4 mm width for cover.

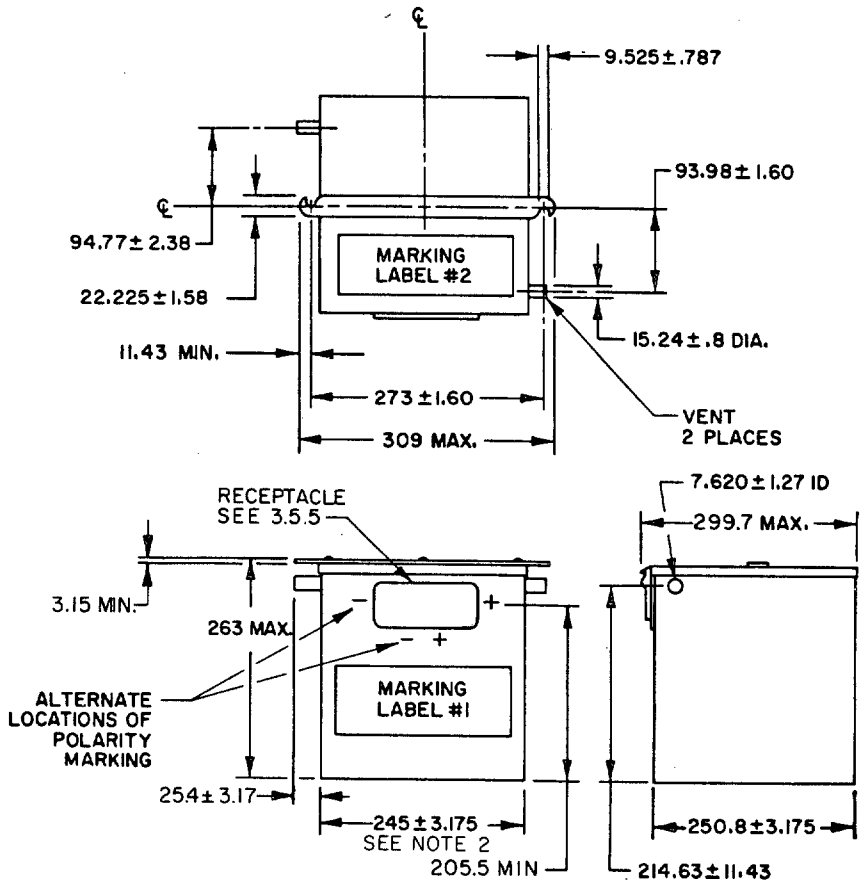
FIGURE 1. Dimensions P/N D8565/5-1.

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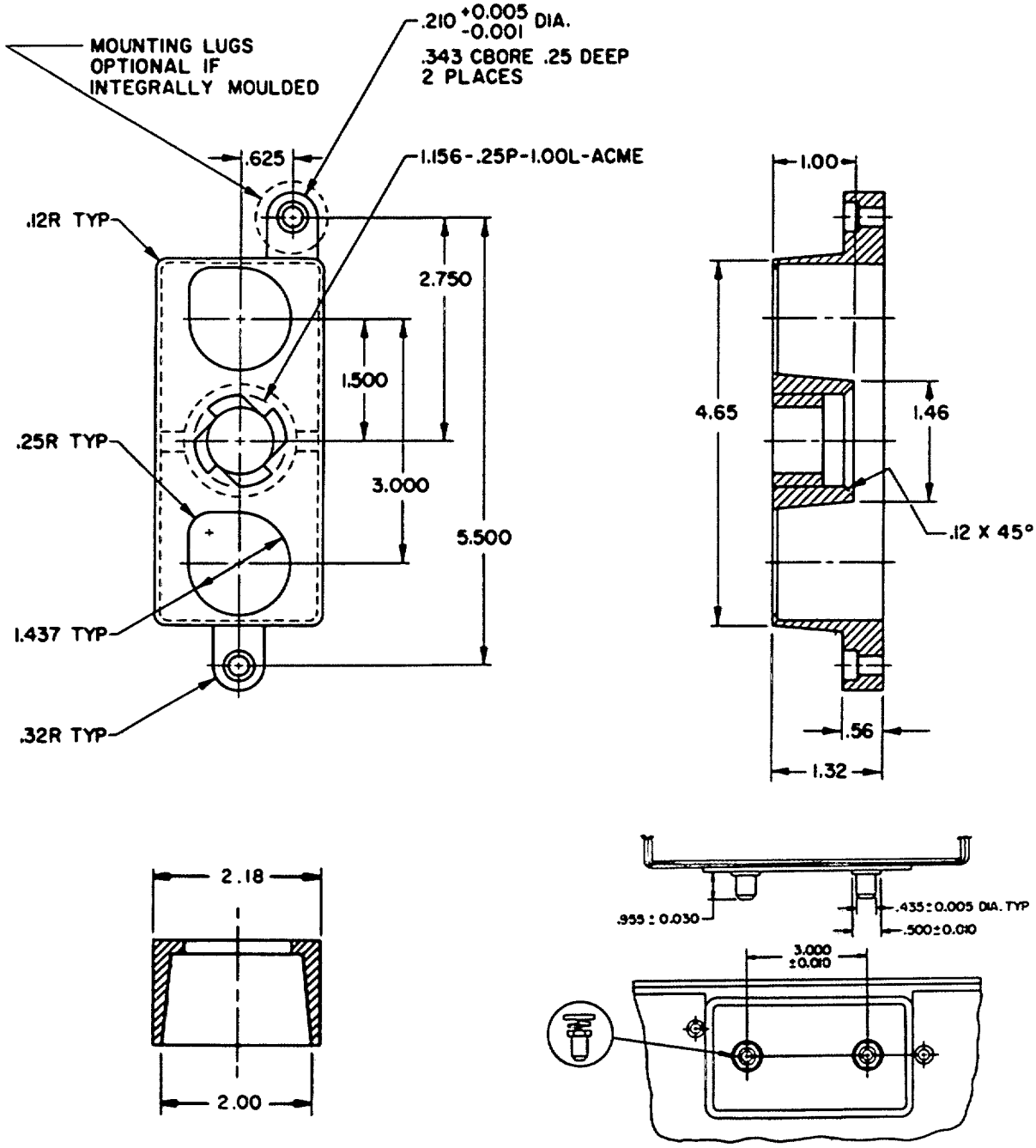



NOTES: 1. All dimensions are in millimeters.
2. Allow additional 4 mm width for cover.

FIGURE 2. Dimensions P/N D8565/5-2.



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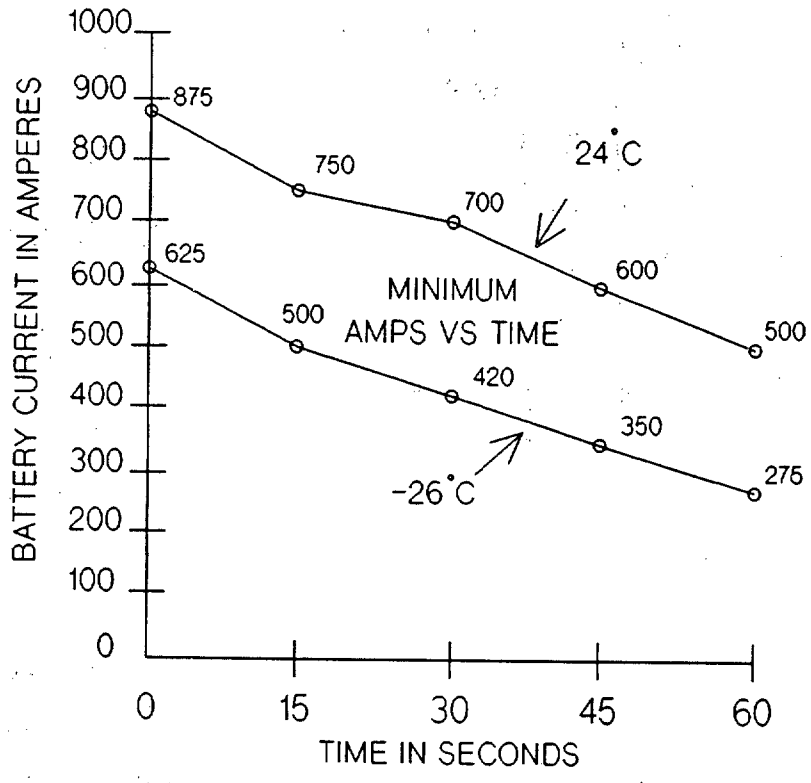



NOTES:
1. Material or finish to be acid-resistant.
2. Dimensions in inches. Tolerances: X.XX (±0.03 in.), X.XXX (±0.010 in.).
3. Receptacle to case attachment method optional for nonmetallic containers.
4. Draft angles allowed in conformance with commercial die casting practice.

FIGURE 3. Typical receptacle adapter.



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| Coordinates of Curves Straight Line Between Coordinates | | |
|---|---|---|
| Time in Seconds | Current in Amperes at | |
| | 24°C | -26°C |
| 0 | 875 | 625 |
| 15 | 750 | 500 |
| 30 | 700 | 420 |
| 45 | 600 | 350 |
| 60 | 500 | 275 |

FIGURE 4. Constant voltage (14.0 V) discharge curves (3.6.8 and 4.6.11).



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REQUIREMENTS:

1. The dimensions of the batteries shall be as shown in Figures 1 and 2.

2. The part number of the battery shall be D8565/5-1 for Figure 1 and D8565/5-2 for Figure 2.

3. The rated capacity of the battery shall be 30.0 ampere-hours (30.0 Ah/1 HR/21°C/18.0 V at end of life). 1C = 30 amperes.

4. The weight of the battery shall not exceed 36.4 kg.

   a. The weight of the batteries supplied under contract, subsequent to qualification, shall not vary by more than ± 1.0 kg from the qualification sample.

5. The battery shall conform to military specification MIL-B-8565, except as modified by the following:

(A) 6. Battery life in the aircraft shall be 3 years.

(A) 3.5.5 Receptacles. Add: "For D8565/5-1 the female receptacle (front side) shall be dimensionally equal to MS3509, style 1. For D8565/5-2 the female receptacle shall be dimensionally equal to Figure 3. The receptacle may be integrally molded in the container."

3.6.8 Capacity and electrical performance. Add the following: "The discharge current shall not drop below the ampere values shown in Figure 4 during the 14.0-volt constant voltage discharge. Inability of the sample to deliver the specified minimum discharge current for 1 minute shall constitute a test failure."

(A) 4.6.13.1 Cycling test. Sentence three, change "2-hour" to "4-hour."

Preparing Activity:
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TABLE 1. Inspection of batteries and order of test.

| Inspection Number | Inspections | Qualification Inspection Qualification Sample Number 1 | 2 | 3 | Requirements Paragraph | Method of Inspection Paragraph | Quality Conformance Inspection Group A Tests | Group B Tests 1 | 2 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Visual & Mechanical Examination | X | X | X | 3.5, 3.6.2.2 & 3.6.3 | 4.6.3 | X | X | X |
| 2 | Dimensions & Weight | X | X | X | 3.6.5 | 4.6.4 | X | X | |
| 3 | Dielectric Strength | X | X | X | 3.6.4 | 4.6.5 | X | X | X |
| 4 | Strength of Vent Tubes | X | X | X | 3.6.7 | 4.6.7 | X | X | X |
| 5 | Color and Marking | X | X | X | 3.6.3 | 4.6.8 | X | X | X |
| 6 | Conditioning Charge | X | X | X | 4.5.1.4 | 4.6.9 | | X | X |
| 7 | Capacity Discharge | X | X | X | 3.6.8 | 4.6.10 | X | X | X |
| 8 | Constant Voltage Discharge | X | X | | 3.6.8 | 4.6.11 | X | X | |
| 9 | Strength of Receptacle | | | X | 3.6.9 | 4.6.12 | | | X |
| 10 | Life Cycling | X | | | 3.6.10 | 4.6.13 | | X | |
| 11 | Capacity Discharge at 49°C | | X | | 3.6.8 | 4.6.14 | | | |
| 12 | Charge & Discharge Test at Low Temperature | | X | | 3.6.8 | 4.6.15 | | | X |
| 13 | Discharge While Inverted | | X | | 3.5.8 & 3.6.8 | 4.6.16 | | | X |
| 14 | Altitude | | X | | 3.6.11 & 3.6.12 | 4.6.17 | | | X |
| 15 | Mechanical Shock | X | | | 3.6.11 & 3.6.12 | 4.6.18 | | | |
| 16 | Temperature Shock | X | | | 3.6.11 & 3.6.12 | 4.6.19 | | X | |
| 17 | Temperature Rise & Float | | X | | 3.6.13 | 4.6.21 | | | X |
| 18 | Battery Gas Emission Test | | X | | 3.6.14 | 4.6.22 | | | |
| 19 | Vibration | | X | | 3.6.16 | 4.6.23 | | | X |
| 20 | Humidity | | X | | 3.6.17 | 4.6.24 | | | X |
| 21 | Salt Fog | X | | | 3.6.18 | 4.6.25 | | X | |
| 22 | Ground Storage | | | X | 3.6.15 | 4.6.26 | | | |
| 23 | Physical Integrity at High Temperature | X | | | 3.6.12 | 4.6.27 | | | |
| 24 | Impact Resistance Test | | 1/ | | 3.6.19 | 4.6.27 | | | |
| 25 | Final Examination | X | X | X | 3.6.4 | 4.6.30 | | X | X |

1/ For nonmetallic battery container designs, two outer nonmetallic battery containers must be supplied. The containers shall be subjected to the impact resistance test (paragraph 4.6.27).

SUPERSESSION DATA. D8565/5-1 supersedes M83769/5-1 and M83769/5-2.
D8565/5-2 supersedes M83769/1-1 and M83769/6-2.





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